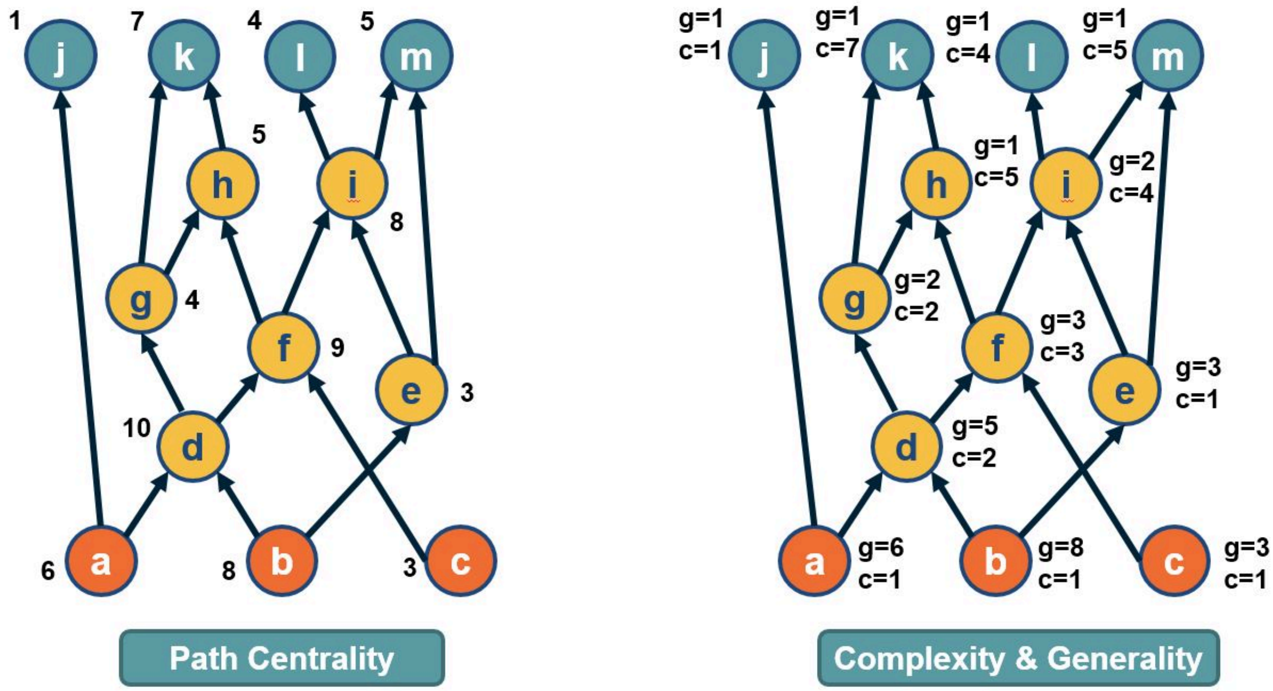
L6: Path Centrality For Directed Acyclic Graphs



In directed acyclic graphs *(DAGs)*, we can consider all paths from the set of sources to the set of targets. The visualization above shows a DAG with three sources *(orange nodes)* and four targets *(blue nodes).* Each source-target path *(ST-path)* represents one ***"dependency chain"*** through which that target depends on the corresponding source.

The path centrality of a node (*including sources or targets) i*s defined as the total number of source-target paths that traverse that node.

It can be easily shown that the path centrality of node-*i* is the product of the number of paths from all sources to node-*i*, times the number of paths from node-*i* to all targets. The former term can be thought of as the ***"complexity"*** of node-*i*, while the second term can be thought of as the ***"generality"*** of node-*i*.